

Pricing Weather Derivatives using Esscher Transforms

SURGE Project Report

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July 2026

1 Introduction

Traditional crop insurance in India is often a very slow and inefficient process. When harsh weather conditions destroy the produce, farmers have to wait for long time to get any compensation for loss. Many times this delay can take months, pushing affected agricultural communities into debt. Weather derivatives offer an alternative solution to this problem. Instead of evaluating the damage on ground level one by one, these financial contracts pay out based on recorded weather data—like cumulative rainfall measured by the India Meteorological Department (IMD). The recent introduction of RAINMUMBAI by NCDEX points to the need of weather derivatives in India.

However, pricing these weather derivatives poses mathematical challenge. Standard methods such as the Black-Scholes assume that asset prices follow normal distribution. If we apply this to the Indian monsoon the model fail completely. Actual rainfall does not move in smooth increments it is characterized by sudden jumps, extended dry spells, and heavy-tailed distributions.

The primary objective of this project is to deduce a method to price these weather derivatives. By replacing normal distributions with Variance Gamma (VG) process, we can capture the sudden extremes of the monsoon. Then we can apply the Esscher Transform to calculate the risk-neutral parameters. Ultimately, this framework provides a reliable method to price climate risk.

2 Methodology

To accurately model the Indian rainfall cannot use standard models like the Brownian motion. Rainfall does not happen in smooth, continuous ticks. Instead we see long dry weather followed by sudden severe bursts of precipitation. To capture this nature of weather we use Variance Gamma (VG) Lévy Process.

The best way to visualize the VG process is to think of it as a standard Brownian motion evaluated on a random Gamma time clock (G). In simple terms, on days with massive rainfall, this internal clock ticks faster than the real-world calendar clock allowing the model to instantly capture this sudden burst. The characteristic function of this process is given by:

$$\phi_{VG}(u) = \left(1 - i\theta\nu u + \frac{1}{2}\sigma^2\nu u^2\right)^{-\frac{1}{\nu}} \quad (1)$$

In this equation, each parameter has a specific physical meaning:

- σ represents the baseline volatility of the weather.
- θ is the drift parameter which controls the overall skewness of the rainfall distribution.
- ν controls the variance of the time change clock. A higher ν means rainfall events are sudden, while a lower ν means a more continuous uniform rainfall.

2.1 Fitting VG-Process to histrocal rainfall dataset

To fit these parameters to real world data we can use a dataset of 25 years of daily rainfall(in mm) of Mumbai Region from the India Meteorological Department (IMD). Using this 25 years of data ensures that the model is based on the true climate conditions.

We calculate the empirical variance (V), skewness (S), and kurtosis (K) from this IMD data using the Method of Moments.

During the winter months when rainfall is almost zero it causes the historical kurtosis to drop. This can result in a negative ν parameter. To fix this we use a floor:

$$\nu = \max\left(0.1, \frac{K - 3}{3.0}\right) \quad (2)$$

This simple bound eliminates the risk of negative ν parameter.

2.2 Payoff Structure & Market Simulation

Before we can price the derivative we need to define its payoff structure. I_{sim} is the average monthly rainfall and K is strike rainfall, the payout is calculated using a non-linear function:

$$\text{Payoff}(I_{sim}) = \alpha \times \max(I_{sim} - K, 0) \quad (3)$$

Here, α is the multiplier factor which is the monetary value given to every millimeter of rainfall which is greater than the strike value K . The max function makes sure that the underwriter or farmer never owes negative money.

If we want to calibrate an pricing model we can simply look at the live prices of options on an exchange. However a liquid market for weather derivative does

not exist in India as of now. Due to this we cannot just take the price from an exchange. To resolve we simulated a market ourselves using the historical physical probability measure.

To do this we used our 25-year rainfall dataset to simulate the market. We calculated the average expected payout for our historical cycles.

However in any real world scenario a Risk-Averse person will never sell a contract for its historical price they will require a margin of safety for the risk. So, we apply a Risk Premium ($\lambda = 0.15$, or a 15% markup). Finally, we will discount this future payout back to get its Day-0 value using a continuous domestic risk-free rate ($r = 0.07$, or 7%).

The resulting simulated market price is given as:

$$P_{\text{market}}(K) = e^{-rT} \times E^P[\text{Payoff}] \times (1 + \lambda) \quad (4)$$

2.3 The use of Esscher Transforms

Because weather derivative market is an incomplete market so we cannot use standard risk neutral replication strategies. To achieve a arbitrage-free valuation in this incomplete market we use the **Esscher Transform**. This is a powerful tool which allows us to change our probability framework from the physical measure (P) to a risk-neutral pricing measure (Q).

The main objective of using Esscher Transforms is to find a esscher risk-aversion constant (h^*) that makes the expected discounted payoff under our risk-neutral measure to behave like a martingale. This condition is represented by the equation:

$$E^Q[\text{Payoff}](h^*) - e^{-rT} E^P[\text{Payoff}](1 + \lambda) = 0 \quad (5)$$

The denominator of the MGF OF VG process under the physical measure is $\kappa(h) = 1 - \theta\nu h - 0.5\sigma^2\nu h^2$. Under Risk neutral parameters for the VG process can be represented in terms of the esscher risk-neutral parameter and physical measure parameters as:

$$\theta^Q = \frac{\theta + \sigma^2 h}{\kappa(h)} \quad (6)$$

$$\sigma^Q = \frac{\sigma}{\sqrt{\kappa(h)}} \quad (7)$$

$$\nu^Q = \nu \quad (8)$$

Notice that the clock variance parameter (ν) remains invariant ($\nu^Q = \nu$) under this change of measure. This is because although the risk premium shifts the expected mean and volatility toward worst-case scenarios, the physical structure of how rainfall cluster together remains identical.

A challenge in this approach is that when we use a standard numerical root-solving algorithm as it tries to find h , it may test a value which forces the

denominator $\kappa(h)$ to be zero. In such a case the risk-neutral drift θ^Q tends toward infinity which forces the model to crash instantaneously.

To address this problem and to make sure that our model doesn't break we introduced a Quadratic Root Bound technique. Before we start the numerical optimizer we firstly solve the underlying quadric equation by making it equal to zero:

$$-0.5\sigma^2\nu h^2 - \theta\nu h + 1 = 0 \quad (9)$$

By solving this using the standard quadratic formula we remove the asymptotic boundaries of h . Then we can fix the numerical root-solver inside the safe 85% range of these boundaries. This constraint will make sure that we never touch a problematic zone and thus it guarantees an accurate and fast convergence.

3 Results

Once we solve for the Esscher risk-neutral parameter for our baseline strike ($K = 0$ mm) this value will act as a constant for that specific month. We can then put h^* back to the Esscher Transform equations to get our risk-neutral parameters (θ^Q and σ^Q).

We can then calculate the final option premium. To do this we execute Monte Carlo simulation. For each month, we generate 50,000 random rainfall paths (I_{sim}). For every single path we calculate the payoff. We then calculate the average of all the 50,000 payouts to find the expected value under the risk-neutral measure and then we can discount it back to get the Day-0 price.

The final calculated parameters and the premiums across all the seasonal segments are listed in Table 1 below.

Month	Mean (mm)	Theta (θ)	Sigma (σ)	Target Q Payout	Final Premium
Jan	0.30	0.0651	0.8719	350.52	348.63
Mar	1.80	0.9193	4.7290	2,068.07	2,056.90
Jun	530.21	184.7814	223.9633	6,09,741.22	6,06,447.39
Jul	1024.49	-237.6392	373.2611	11,78,167.82	11,71,803.33
Aug	542.20	447.6806	216.9591	6,23,528.34	6,20,160.03
Oct	67.35	151.9946	47.1130	77,455.18	77,036.77

Table 1: Calculated Weather Options Prices via VG-Esscher method

These results prove that the method works exactly as intended. The model successfully gave the fair-value of premiums for both volatile peak monsoon conditions (June, July, August) and dry winter transitions (January, March). The entire 50,000-path options chain was generated without a single computational crash, ensuring the method remains robust under real-world loads.

4 Socio-Economic Impact

While the mathematical methodology relies on stochastic calculus and risk-neutral transformations, the main goal of this project is very practical. This framework is built to solve a critical, real-world problem which is the vulnerability of India's agricultural sector to extreme climate shocks.

More than half of India's workforce depends directly on the rainfall cycle. Under the current scenario if a crop fails due to harsh weather conditions the farmers suffer the physical loss immediately but they have to wait for months to get any sort of compensation. This delay pushes smallholder farmers into the vicious cycle of debt traps.

This pricing method will allow the regional cooperative banks to structure arbitrage-free Over-The-Counter (OTC) insurance. By pooling resources to pay these calculated premiums, Farmer Producer Organizations can transfer their weather risk directly to institutional capital markets.

Because the derivative is parametric, payouts trigger the exact moment a rainfall deficit is recorded by the IMD—no manual field inspections required. This immediate cash injection means a farmer can quickly purchase seeds for a secondary, drought-resistant crop rather than losing the entire season to a single failed monsoon.